A Two to six overlapping exposures, in the order a radiographer already works

same field size throughout; each frame keeps two bodies of the previous one in view

overlap = 2 bodies

encoder
ResNet-18, one crop at a time

one embedding per vertebra

B The same embeddings, named two ways

each crop judged alone
argmax over 27 classes
70.7%

two vertebrae in view, no landmark in the pixels

linked into one chain
frames joined on the bodies they share

its vertebrae are consecutive, and its lower end is the sacrum

C Where the ordering constraint enters, and what it is worth

What the chain supplies
that its vertebrae are consecutive,
and where the run ends.
Neither fact is in the pixels
of a limited field.

sequence head
one name for the whole run

chain decode
four alphabets scored

93.4%
the same frame,
named on the chain

ordinary L6 T13 both

a sixth lumbar vertebra is a different ALPHABET, never a rare class

+22.7 points, on the same pixels

n = 801 held-out spines, 3,222 two-vertebra frames